5.12.4 Complementary Symmetry Amplifier

Complementary Symmetry is based on the principle of assembling a push-pull class B amplifier without requiring centre-tapped transformers, at the input and output stages.

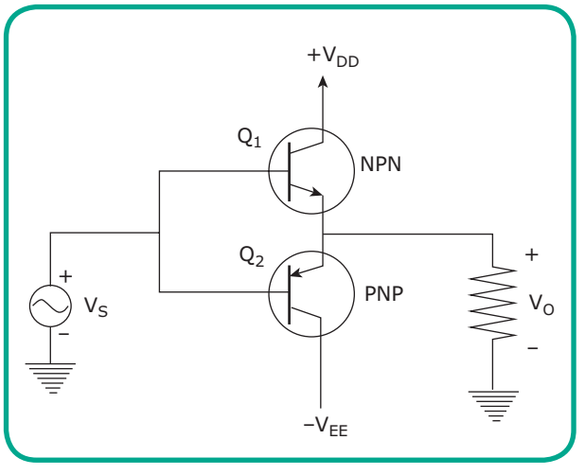


Figure 5.29 Simplified complementary symmetry amplifier

Figure 5.29 shows the transistor push-pull amplifier using complementary symmetry. It employs one NPN and one PNP transistor and requires no centre-tapped transformers. The circuit action is follows. During the positive half cycle of the input signal, transistor Q_1(the NPN transistor) conducts current while Q_2(the PNP transistor) is cut-off. During the negative half cycle of the input signal, transistor Q_2 (the PNP transistor) conducts current while Q_1 (the NPN transistor) is cut-off.

In this way, NPN transistor amplifies the positive half cycles of the signal while the PNP transistor amplifies the negative half cycles of the signal. Note that we generally use an output transformer (not centre tapped) for impedance matching.

Advantages

1. This circuit does not require transformer. It reduces both weight and cost.
2. Equal and opposite input signal voltages are not required.

Disadvantages

1. It is difficult to get a pair of transistors (NPN and PNP) having similar characteristics.
2. It requires both positive and negative supply voltages.

Application

This type of amplifier is used in stereo amplifiers.

5.12.5 Differential Amplifier

A differential amplifier amplifies the voltage difference between the two inputs. An example of a configuration of the amplifier is connecting the emitters of two transistors with equal characteristics as shown in Figure 5.30(a). The voltage difference applied to the base of these transistors is amplified. Figure 5.30(b) shows the waveforms at each node of the differential amplifier.

The output of differential amplifier circuitry shown in Figure 5.30(a) is further amplified and applied utilizing negative feedback. Presently, high performance and inexpensive differential amplifiers, so-called operational amplifiers (integrated circuit), are available in the market.

If two input signals of the differential amplifier are the same, the (differential) output will be nearly zero. That is, common mode noise of the input is greatly attenuated, and only the signal is amplified. A function of the amplifier to reduce the common-mode noise is CMRR (common- mode rejection ratio).

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